SeaCube: An Open-Source Autonomous Docking Station For Long Endurance, Unattended AUV Operations

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Software Control Architecture

A Raspberry Pi with an handles mission logic, data buffering, and inference at the dock, using ROS2. while a Teensy 4.1 manages real-time actuation and sensor polling. A contact limit switch and IR proximity sensor communicate to the station when a vehicle is aboard independently of the radio, and can gate the charging transmitter. Arriving vehicles home in on adaptive-color LED markers that recompute their hue from the surrounding water, holding contrast as visibility shifts [1]. This dual occupancy check allows for the dock to work under broken radio communications making the platform perform reliably in the field.

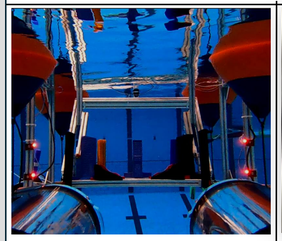


Fig. 1: Light based visual docking for AUVs [1].



Fig. 2: Physical layout of the electronics present in SeaCube.

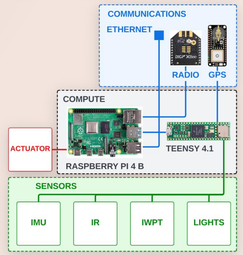
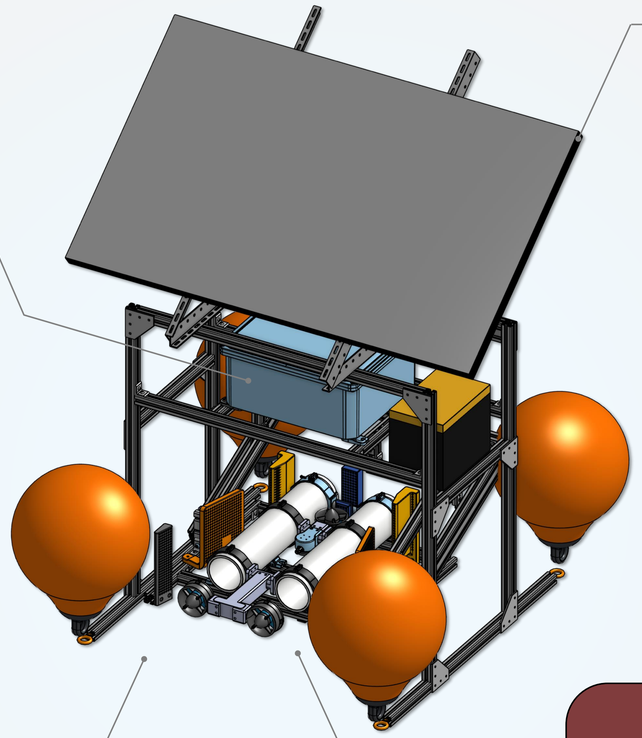


Fig. 3: Communication layout of electronics in SeaCube.

- SeaCube is a **sustainable solar powered**, **AI-capable**, remote AUV service station
- It costs **$2000** in off-the-shelf and printed parts,and is released **open-source**
- It is a platform towards **sustainable**, **AI-powered robotic environmental monitoring** and **conservation**



Solar Power Assembly

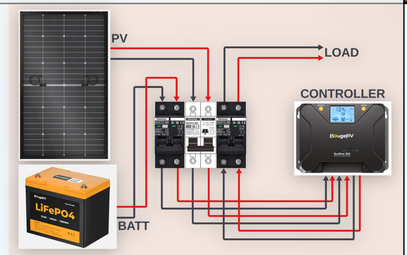


Fig. 4: Schematic of the solar power harvesting system onboard SeaCube.

- **130 W of surplus power**, measured across field tests, enough for charging and onboard compute both
- **83%** of an AUV battery returned overnight on stored energy alone
- **200 W bifacial panel**, **100 Ah LiFePO4** bank, MPPT controller, chosen for passing clouds on open water

A 200 W bifacial panel at 40° tilt charges a 100 Ah LiFePO4 bank through an MPPT controller. Three 30 A breakers protect the panel, the battery, and everything downstream. The station draws 12.2 W against 145–153 W generated, leaving a 130 W surplus across both deployments.

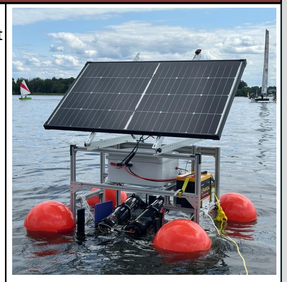


Fig. 5: SeaCube deployed in lake Minnetonka for field testing in July.

Parameter	Phase 1		Phase 2	
	Undocked	Docked	Undocked	Docked
PV Voltage (V)	29.20	-	30.40	29.20
PV Current (A)	4.96	-	5.03	5.10
PV Power (W)	144.83	-	152.91	148.92
Battery Voltage (V)	13.50	-	13.50	13.50
Battery Current (A)	10.32	-	10.88	10.59
Battery Power (W)	139.32	-	146.88	142.97
Load Voltage (V)	13.40	-	13.50	13.50
Load Current (A)	1.00	-	0.90	1.42
Load Power (W)	13.40	-	12.15	19.17
Temperature (°C)	25.00	-	25.00	25.00

Table 1: Power comsumption metrics drawn from two field tests spanning across 7 hours in duration

Radio Communications and GPS

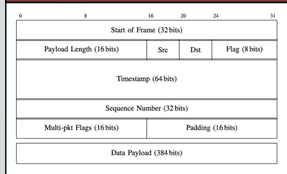
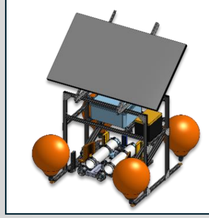


Fig. 6: Diagram of the custom message protocol (above).

An XBee 3 Pro running DigiMesh forms a network with no coordinator node, so a single AUV holding a link can relay for the rest of a fleet. A custom 72-byte frame stays usable down to 1 kbit/s, which matters at the 0.34 kbit/s seen at 50 m with only 0.2 m of antenna above the waterline. Bulk transfers after the robot is docked, where throughput rises to 100–250 kbit/s. The station also broadcasts its own drifting GPS position, accurate to 1.8 m, to vehicles on approach.



Fig. 8: Field testing at Square lake in August.



Bidirectional Radio



Fig. 7: Radio performance measured in kbit/s (right).

Distance (m)	Mean	Std. dev. (kbit/s)	Min	Max
5	6.50	1.00	3.45	7.68
10	4.26	1.43	2.61	6.39
20	6.32	0.07	6.11	6.48
40	1.17	0.61	0.05	2.22
50	0.34	0.13	0.06	0.60
60	0.00	0.00	0.00	0.00

Cradle Mechanism

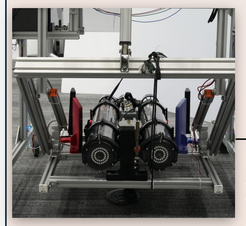
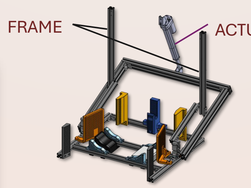


Fig. 9: LoCO AUV [2] docked in the Cradle on SeaCube.



Two closed linkages driven by a 12 V, 2000 N actuator lift the AUV clear of the water with its orientation preserved, so servicing happens dry. Printed guides and seats converge the vehicle onto a catch rod that completes the dock.

Inductive Wireless Power Transfer (IWPT)

A 138 kHz module transfers power across coils curved to match the AUV's hull, separated by 17.3 mm of acrylic and printed enclosure wall. Docking draws 7.5 W from the station bus, a figure confirmed independently at the charge controller. Testing under two receiver loads an order of magnitude apart shows the increase tracking the load rather than mere proximity, which establishes that this is genuine coupling. The gap limits the rate, not the power budget: a thinner transmitter case or a 150 W-class module can speed up charning greatly. Results can be seen in the graphs below.

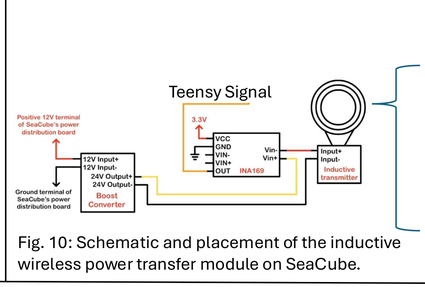
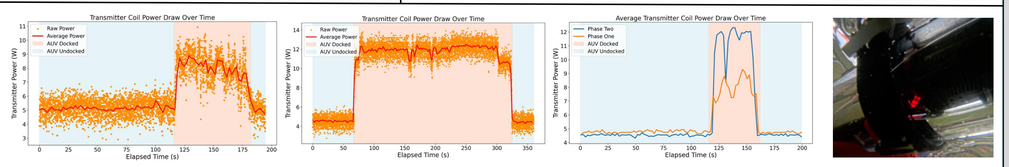


Fig. 10: Schematic and placement of the inductive wireless power transfer module on SeaCube.



References:

[1] - C. Knutson, Z. Cao and J. Sattar, "Adaptive Landmark Color for AUV Docking in Visually Dynamic Environments," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 9990-9996.

[2] - C. Edge et al., "Design and Experiments with LoCO AUV: A Low Cost Open-Source Autonomous Underwater Vehicle," 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Las Vegas, NV, USA, 2020, pp. 1761-1768,

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